

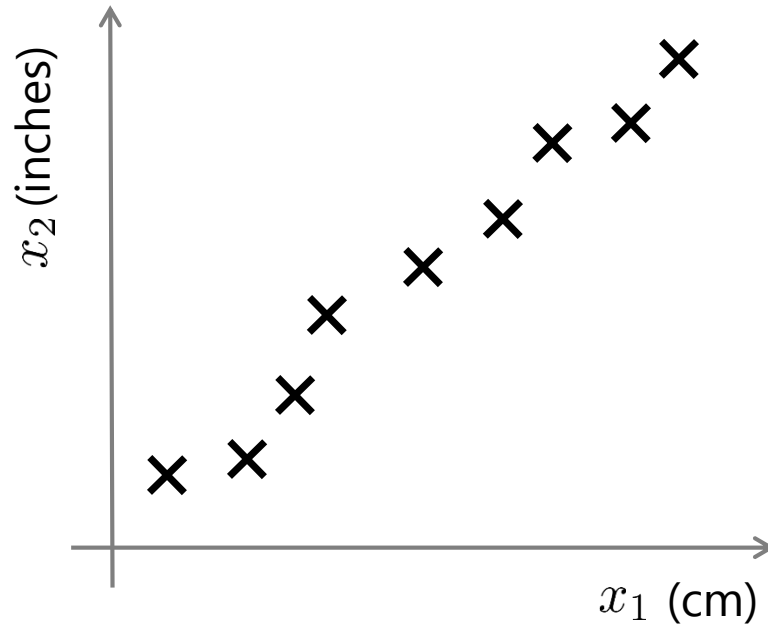
Dimensionality Reduction

Taweesak Samanchuen

Motivation I:

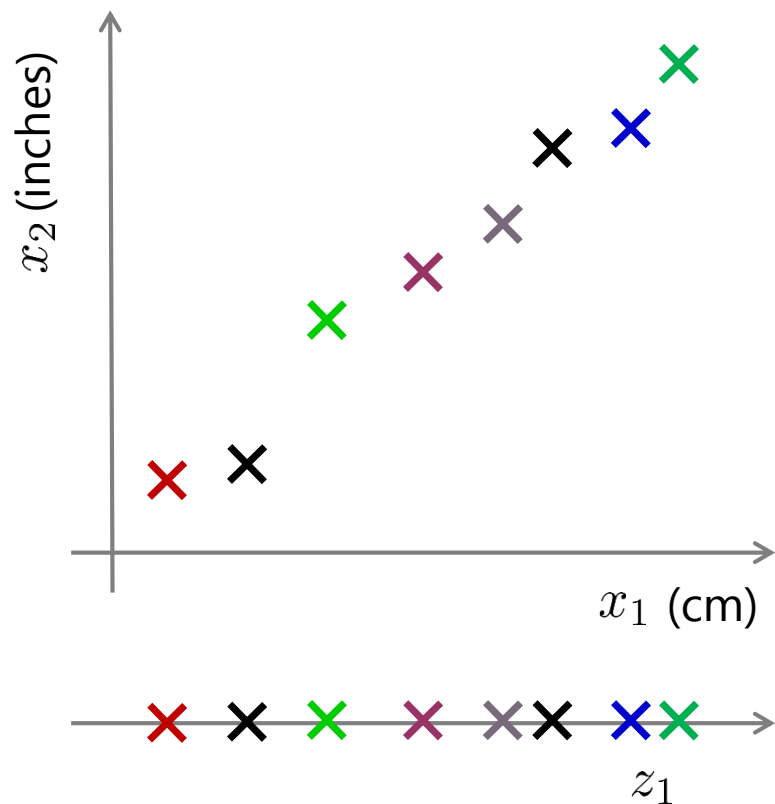
Data Compression

Data Compression



Reduce data from
2D to 1D

Data Compression



Reduce data from
2D to 1D

$$x^{(1)} \rightarrow z^{(1)}$$

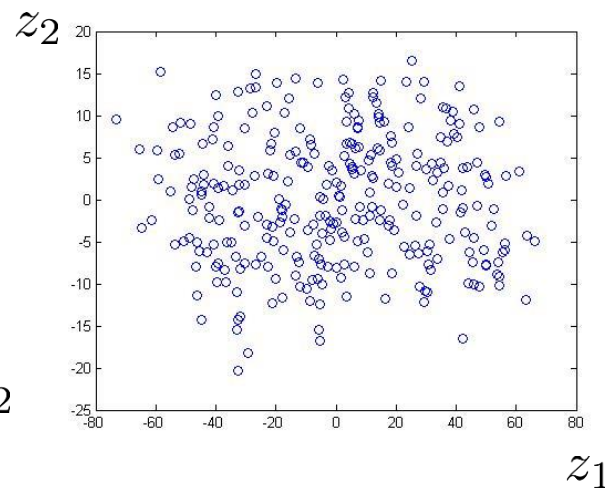
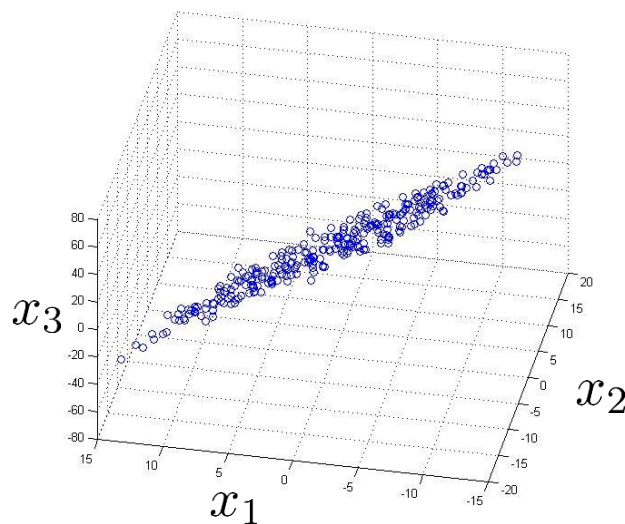
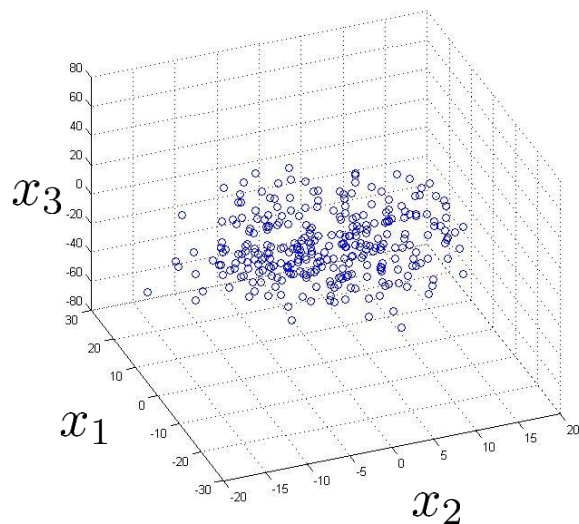
$$x^{(2)} \rightarrow z^{(2)}$$

$\vdots$

$$x^{(m)} \rightarrow z^{(m)}$$

Data Compression

Reduce data from 3D to 2D



Motivation II:

Data Visualization

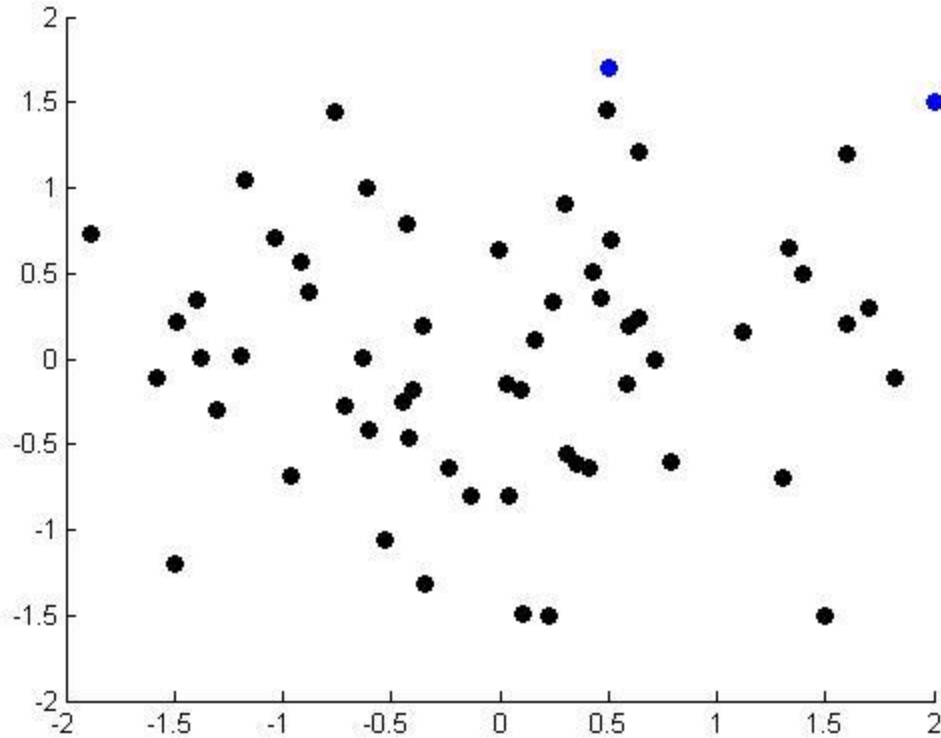
Data Visualization

Country	GDP (trillions of US\$)	Per capita GDP (thousands of intl. \$)	Human Development Index	Life expectancy	Poverty Index (Gini as percentage)	Mean household income (thousands of US\$)	...
Canada	1.577	39.17	0.908	80.7	32.6	67.293	...
China	5.878	7.54	0.687	73	46.9	10.22	...
India	1.632	3.41	0.547	64.7	36.8	0.735	...
Russia	1.48	19.84	0.755	65.5	39.9	0.72	...
Singapore	0.223	56.69	0.866	80	42.5	67.1	...
USA	14.527	46.86	0.91	78.3	40.8	84.3	...
...	...	...	...	...	...	...	...

Data Visualization

Country	z_1	z_2
Canada	1.6	1.2
China	1.7	0.3
India	1.6	0.2
Russia	1.4	0.5
Singapore	0.5	1.7
USA	2	1.5
...	...	...

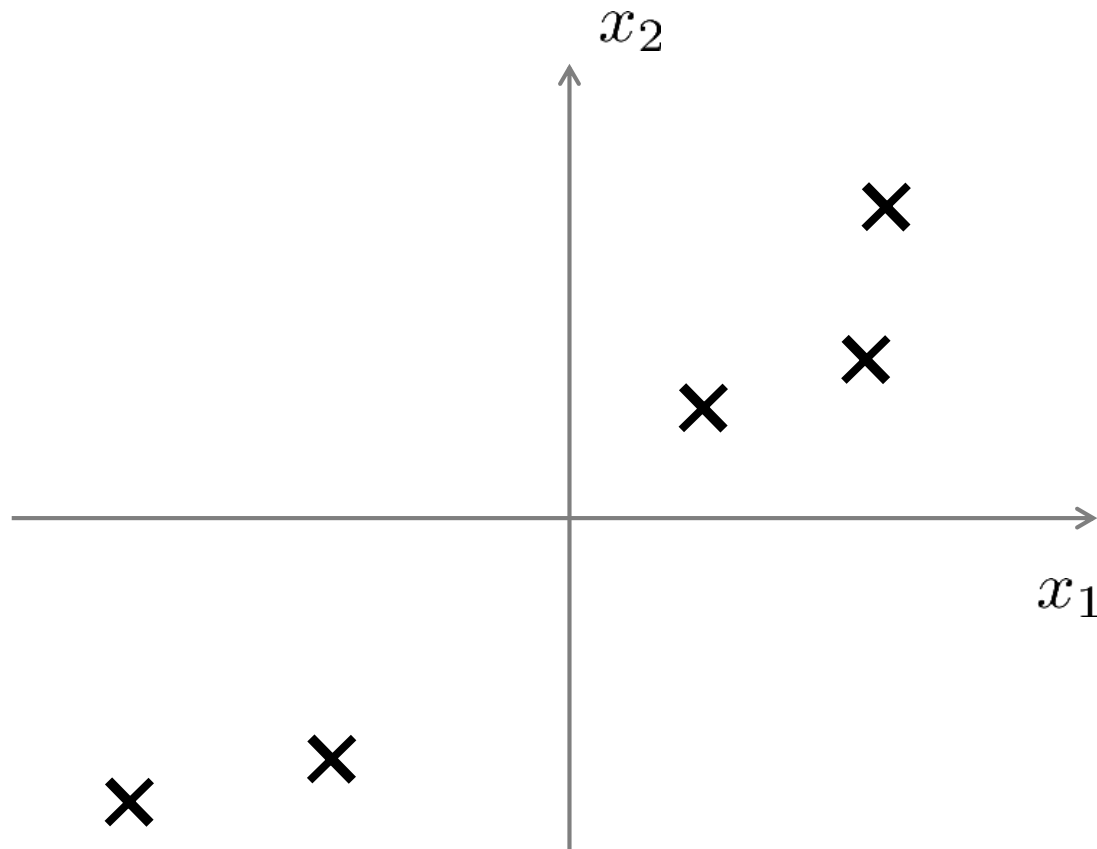
Data Visualization



Problem formulation

Principal Component Analysis

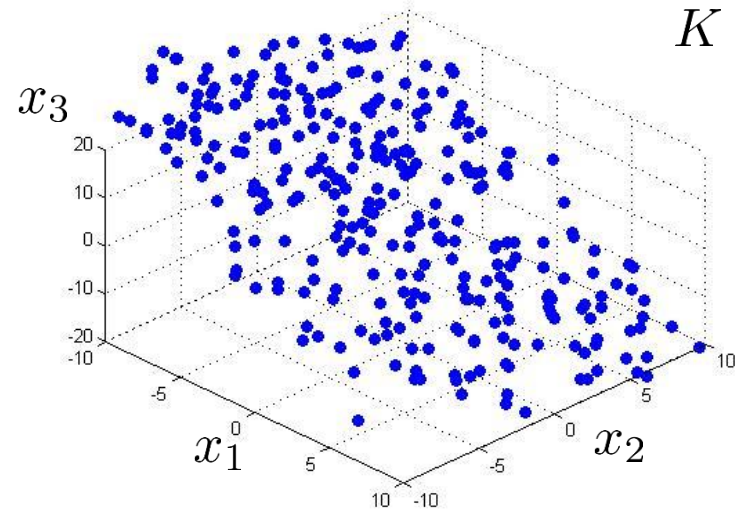
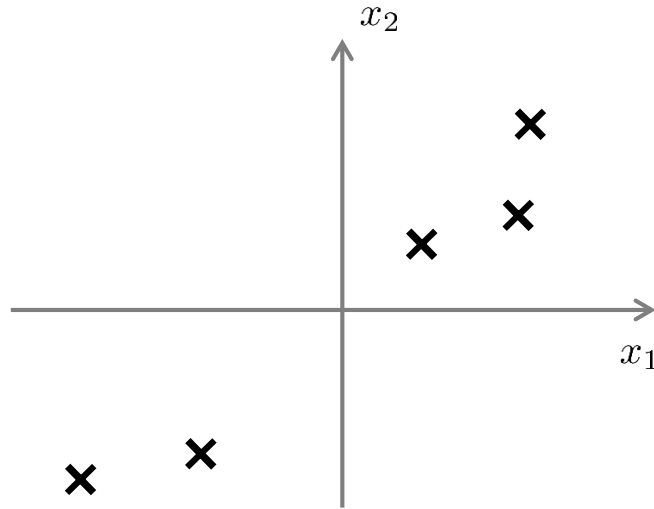
Principal Component Analysis (PCA) problem formulation



Principal Component Analysis (PCA) problem formulation

$$3D \rightarrow 2D$$

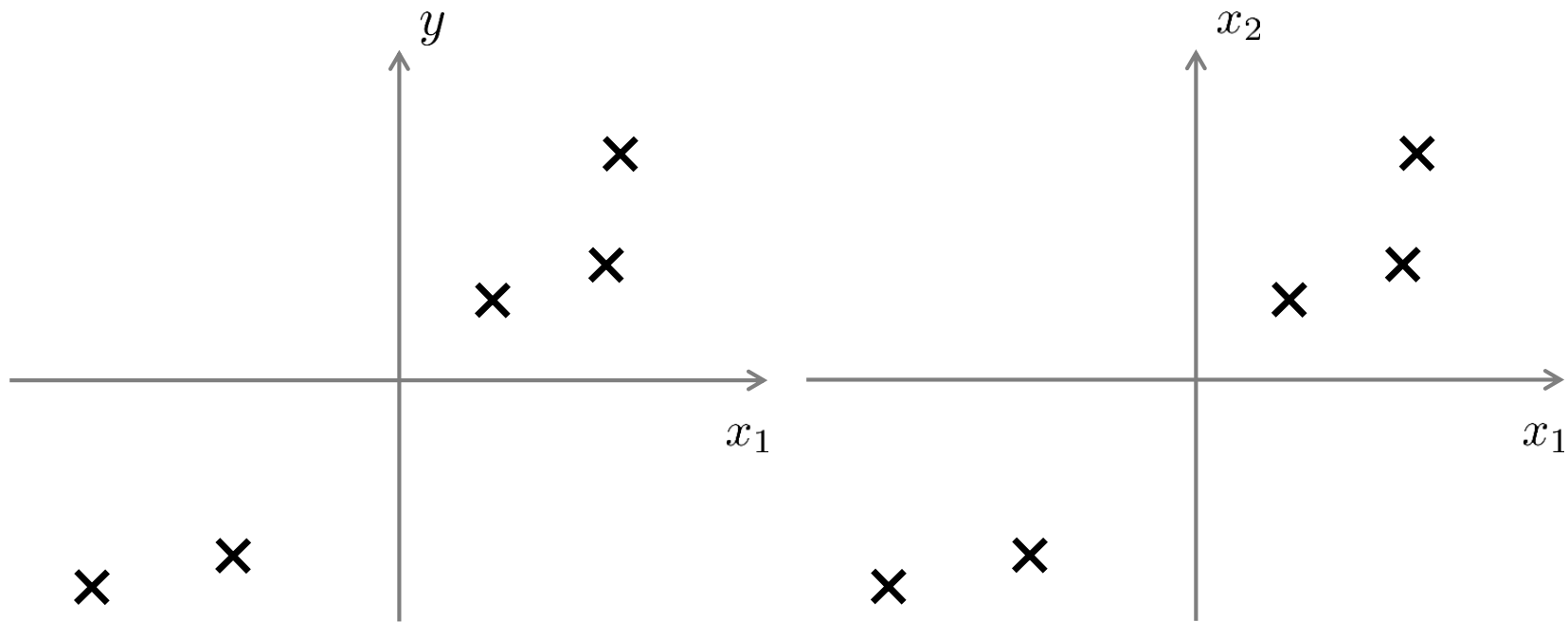
$$K = 2$$



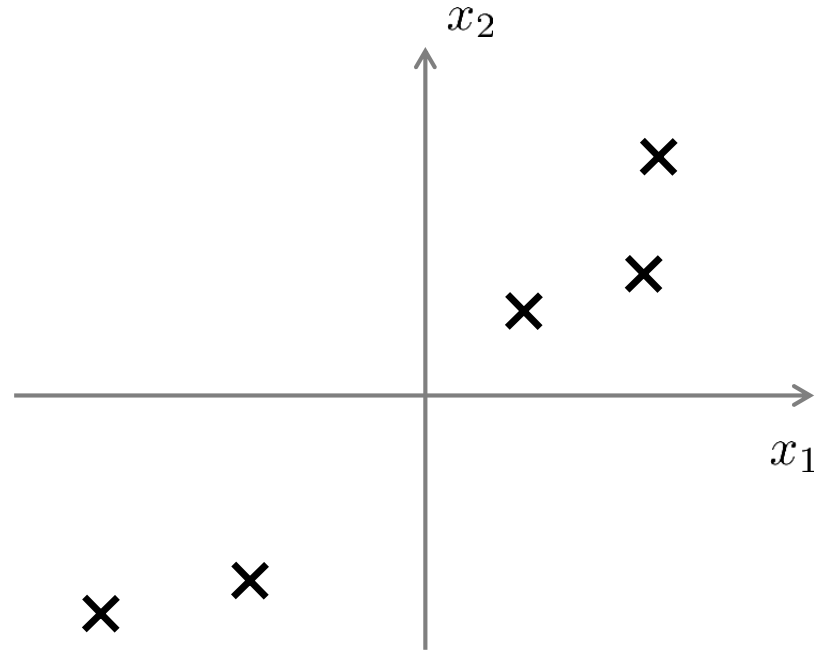
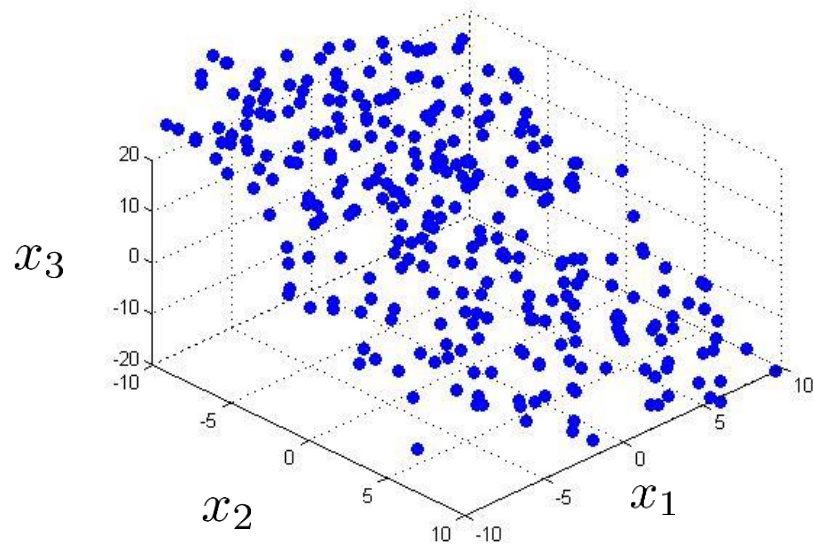
Reduce from 2-dimension to 1-dimension: Find a direction (a vector $\mathbf{v}^{(1)} \in \mathbb{R}^n$) onto which to project the data so as to minimize the projection error.

Reduce from n-dimension to k-dimension: Find vectors $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(k)}$ onto which to project the data, so as to minimize the projection error.

PCA is not linear regression



PCA is not linear regression



Algorithm

Principal Component Analysis

Data preprocessing

Training set: $x^{(1)}, x^{(2)}, \dots, x^{(m)}$

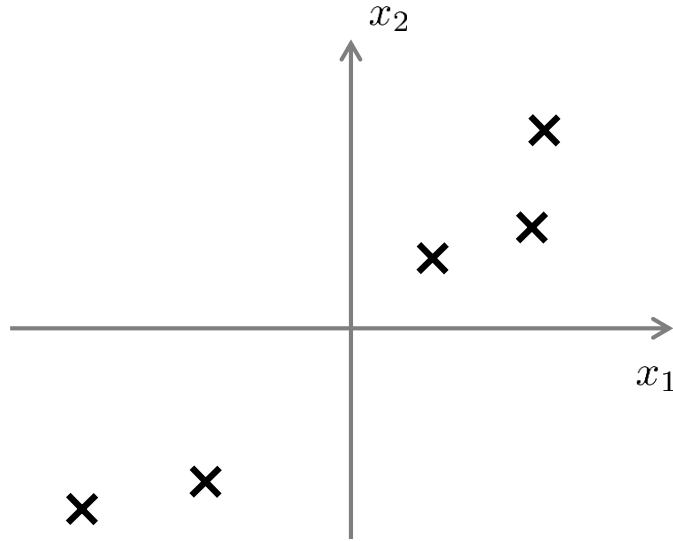
Preprocessing (feature scaling/mean normalization):

$$\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)}$$

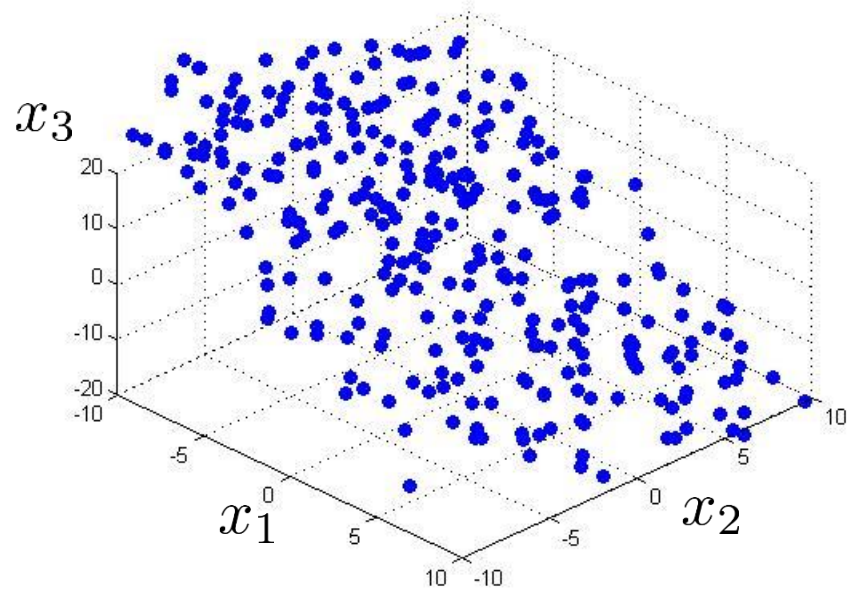
Replace each $x_j^{(i)}$ with $x_j - \mu_j$.

If different features on different scales (e.g., x_1 = size of house, x_2 = number of bedrooms), scale features to have comparable range of values.

Principal Component Analysis (PCA) algorithm



Reduce data from 2D to 1D



Reduce data from 3D to 2D

Principal Component Analysis (PCA) algorithm

Reduce data from m -dimensions to k -dimensions

Compute "covariance matrix":

$$\Sigma = \frac{1}{m} \sum_{i=1}^n (x^{(i)})(x^{(i)})^T$$

Compute "eigenvectors" of matrix Σ :

`[U,S,V] = svd(Sigma);`

Principal Component Analysis (PCA) algorithm

From $[U, S, V] = \text{svd}(\text{Sigma})$, we get:

$$U = \begin{bmatrix} | & | & \dots & | \\ u^{(1)} & u^{(2)} & & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

Principal Component Analysis (PCA) algorithm summary

After mean normalization (ensure every feature has zero mean) and optionally feature scaling:

$$\mathbf{Sigma} = \frac{1}{m} \sum_{i=1}^m (x^{(i)})(x^{(i)})^T$$

```
[U,S,V] = svd(Sigma);
```

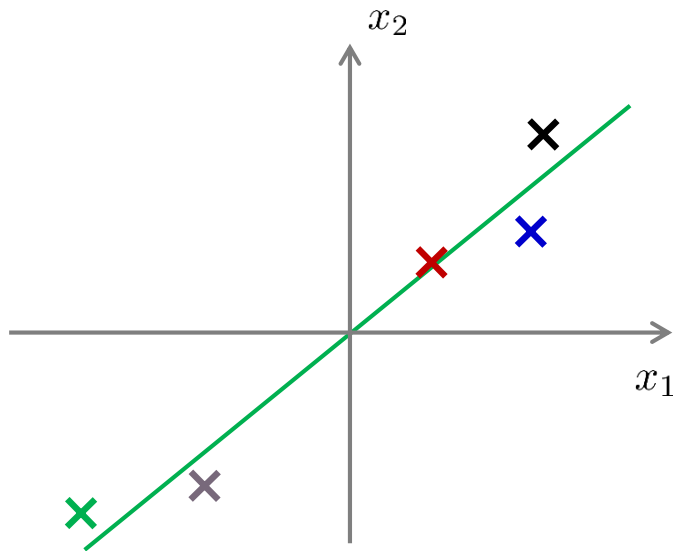
```
Ureduce = U(:,1:k);
```

```
z = Ureduce'*x;
```

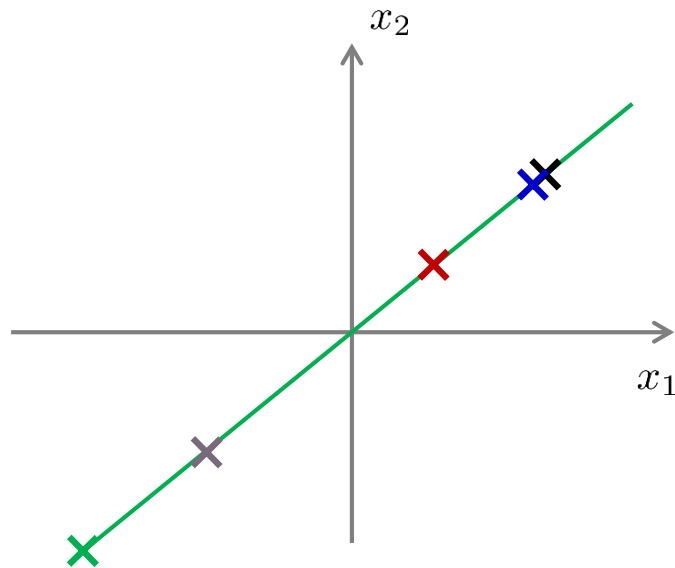
Dimensionality Reduction

Reconstruction from compressed representation

Reconstruction from compressed representation



$$z = U_{reduce}^T x$$



Dimensionality Reduction

Choosing the number of principal components

Choosing k (number of principal components)

Average squared projection error:

Total variation in the data:

Typically, choose k to be smallest value so that

$$\frac{\frac{1}{m} \sum_{i=1}^m \|x^{(i)} - x_{approx}^{(i)}\|^2}{\frac{1}{m} \sum_{i=1}^m \|x^{(i)}\|^2} \leq 0.01 \quad (1\%)$$

"99% of variance is retained"

Choosing k (number of principal components)

Algorithm:

Try PCA with $k = 1$

Compute $U_{reduce}, z^{(1)}, z^{(2)}, \dots, z^{(m)}, x_{approx}^{(1)}, \dots, x_{approx}^{(m)}$

Check if

$$\frac{\frac{1}{m} \sum_{i=1}^m \|x^{(i)} - x_{approx}^{(i)}\|^2}{\frac{1}{m} \sum_{i=1}^m \|x^{(i)}\|^2} \leq 0.01?$$

$$[U, S, V] = \text{svd}(\text{Sigma})$$

Choosing k (number of principal components)

`[U,S,V] = svd(Sigma)`

Pick smallest value of k for which

$$\frac{\sum_{i=1}^k S_{ii}}{\sum_{i=1}^m S_{ii}} \geq 0.99$$

(99% of variance retained)

Dimensionality Reduction

Advice for applying PCA

Supervised learning speedup

$$(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})$$

Extract inputs:

$$\text{Unlabeled dataset: } x^{(1)}, x^{(2)}, \dots, x^{(m)} \in \mathbb{R}^{10000}$$

$$\downarrow \text{PCA}$$

$$z^{(1)}, z^{(2)}, \dots, z^{(m)} \in \mathbb{R}^{1000}$$

New training set:

$$(z^{(1)}, y^{(1)}), (z^{(2)}, y^{(2)}), \dots, (z^{(m)}, y^{(m)})$$

Note: Mapping $x^{(i)} \rightarrow z^{(i)}$ should be defined by running PCA only on the training set. This mapping can be applied as well to the examples $x_{cv}^{(i)}$ and $x_{test}^{(i)}$ in the cross validation and test sets.

Application of PCA



- Compression
 - Reduce memory/disk needed to store data
 - Speed up learning algorithm
- Visualization

Bad use of PCA: To prevent overfitting

Use $z^{(i)}$ instead of $x^{(i)}$ to reduce the number of features to $k < n$.

Thus, fewer features, less likely to overfit.

This might work OK, but isn't a good way to address overfitting. Use regularization instead.

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

PCA is sometimes used where it shouldn't be

Design of ML system:

- Get training set $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$
- Run PCA to reduce $x^{(i)}$ in dimension to get $z^{(i)}$
- Train logistic regression on $\{(z^{(1)}, y^{(1)}), \dots, (z^{(m)}, y^{(m)})\}$
- Test on test set: Map $x_{test}^{(i)}$ to $z_{test}^{(i)}$. Run $h_{\theta}(z)$ on $\{(z_{test}^{(1)}, y_{test}^{(1)}), \dots, (z_{test}^{(m)}, y_{test}^{(m)})\}$

How about doing the whole thing without using PCA?

Before implementing PCA, first try running whatever you want to do with the original/raw data $x^{(i)}$. Only if that doesn't do what you want, then implement PCA and consider using $z^{(i)}$.